InfoSec PreMid Quiz Name: ___________________ Roll No:______________ Date: April 1st, 2026
Marks: 30

1. Which of the following is not a symmetric encryption cipher?

A. AES
B. DES
C. RSA
D. Playfair

2. Which pillar of the CIA Triad is compromised by a DDoS attack?

A. Confidentiality
B. Availability
C. Integrity
D. Non-repudiation

3. Ciphertext: BMNDZBXDKYBEJVDMUIXMMNUVIF, Key: SECURITY, Cipher name is Playfair. Plaintext is

A. DEFENDTHEWESTWALLOFTHECASTLE
B. ATTACKTHEWESTWALLOFTHECASTLE
C. ATTACKTHEEASTWALLOFTHECASTLE
D. DEFENDTHEEASTWALLOFTHECASTLE

4. Which protocol creates a bidirectional channel for continuous data transfer and ensures packets arrive in order?

A. UDP
B. TCP
C. IP (Internet Protocol)
D. HTTP

5. Which of the following can prevent a MITM attack in Diffie-Hellman?

A. Using larger prime (p)
B. Using XOR instead of mod operations
C. Authenticating public keys with digital signatures
D. Using symmetric encryption

6. Which of the following best describes public-key cryptography?

A. Both keys are kept secret, one is shared securely
B. One key is public, the other is secret
C. One Key is shared over secure channels
D. Keys must be the same length

7. In asymmetric encryption, if Bob wants to send a private message to Alice, he uses:

A. Alice's Public key
B. Alice's Private Key
C. Bob's Private Key
D. Bob's Public Key

8. Alice and Bob use Diffie-Hellman and later encrypt messages with AES using the derived key. Oscar records all messages but never interferes. Later, Oscar steals Bob's private key.
What can Oscar do now?

A. Decrypt all past messages
B. Nothing
C. Decrypt only future messages
D. Both past and future messages

9. Before encrypting with Playfair cipher, the message "LETTER" is divided into digraphs. What is the correct prepared form?

A. LE TT ER
B. LE TX TE R
C. LE TX TE RX
D. LE TE TR

10. Which of the following is listed as a countermeasure for confidentiality?

A. Data replication
B. Adequate bandwidth
C. Access control
D. Regular backup

11. **Which Wireshark display filter would you use to show only traffic coming from the IP 192.168.1.10?**

A. ip.addr == 192.168.1.10
B. http.request
C. ip.src == 192.168.1.10
D. tcp.port == 192.168.1.10

12. What is the block size of the Data Encryption Standard (DES)?

A. 56 bits
B. 128 bits
C. 64 bits
D. 192 bits

13. Which OSI layer is responsible for IP address spoofing and routing table poisoning?

A. Data Link Layer
B. Transport Layer
C. Network Layer
D. Application Layer

14. Stuxnet is a famous example of which type of cyber attack?

A. Phishing
B. DDoS
C. Malware
D. SQL Injection

15. According to Asymmetric Cryptography, a sender can digitally sign a message by using which key?

A. The recipient's public key
B. The senders' private key
C. The recipient's private key
D. The sender's public key

Short Answer is required

1. **What is meant by zero-day exploits.**

   A zero-day exploit is an attack that targets a software vulnerability previously unknown to the vendor or the public. The name refers to the fact that developers have had "zero days" to fix the issue.

2. **Difference between Encoding and Encryption.**
   Encoding is the process of transforming data into a specific format for efficient transmission or storage (no secret key involved). Encryption is the process of transforming data into an unreadable format using a secret key to ensure confidentiality.

3. **What is the fundamental Challenge of Symmetric Cryptography that led to the development of Asymmetric Cryptography?**

   The Key Distribution Problem. In symmetric cryptography, parties need a secure way to share the single secret key before they can communicate; if the key is intercepted during sharing, the system is compromised.

4. **Difference between Capture Filter and Display Filter.**
   A Capture Filter is applied before recording traffic (decides what is saved to memory).
   E.g: host 192.168.1.1 Capture only traffic to/from this IP
   A Display Filter is applied after traffic is captured (decides what is shown on the screen).

E.g: ip.addr == 192.168.1.1 Show packets with this IP

5. **Security monitoring systems have identified one malicious IP address: 203.0.113.50. How can a firewall rule be written so that all traffic to and from this IP is blocked?**

1. Block all incoming traffic FROM the malicious IP
iptables -A INPUT -s 203.0.113.50 -j DROP

2. Block all outgoing traffic TO the malicious IP
iptables -A OUTPUT -d 203.0.113.50 -j DROP

Q. Alice and Bob decide to use the Diffie-Hellman key exchange to establish a secret key over an insecure network. They agree on the public parameters p = 29 (a prime number) and g = 2 (a primitive root). Alice chooses her private secret a=5 and Bob chooses his private secret b=12**.**

1. Calculate Alice's public value and Bob's public value.
2. Calculate Shared Secret Key that both Alice and Bob compute independently.
3. Is the DH Key Computation vulnerable to MITM attack.

Alice's Public Value = 3
Bob's Public Value = 7
Shared Secret Key = 16

Vulnerability to MITM attack: Yes. The basic Diffie-Hellman protocol is vulnerable to Man-in-the-Middle (MITM) attacks because it is unauthenticated (Alice knows she received a public value, but she has no way to prove it actually came from Bob). An attacker can intercept the public values and substitute their own, establishing separate secret keys with both Alice and Bob without them knowing.